Loan Default Prediction

Comparing Classic ML and Neural Networks under Class Imbalance

Minwoo Yoo · Nathaniel Badalov

Home Credit Default Risk · Kaggle

Problem & Motivation

The task

Predict the probability that a loan applicant defaults (binary classification, $\texttt{TARGET} \in \{0, 1\}$).

Why this matters

- Lenders need accurate risk scoring to balance approval volume and credit losses.
- False negatives (approving a bad loan) are far more costly than false positives.

Class imbalance challenge

- Only $\sim$**8.07%** of applicants default — the data is severely imbalanced.
- A naive "always predict no default" baseline scores 92% accuracy with **zero recall**.
- Evaluating with accuracy alone is misleading; we use **ROC AUC** as the

Dataset · Home Credit Default Risk

Size

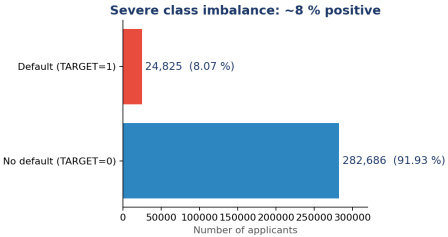
- Train: **307,511** applicants $\times$ 122 features
- Test: 48,744 applicants (Kaggle hidden labels)

Class imbalance

- Default rate $\approx$ **8.07%** (24,825 / 307,511)

Strongest signals

- `EXT_SOURCE_1/2/3` external credit scores ($|\rho| \approx 0.16$–0.18)
- `DAYS_EMPLOYED` = 365,243 sentinel flags 18% (unemployed)
- Building-info columns are 50–70% missing



Shared Pipeline

Preprocessing (identical for all five models)

- **80/20 stratified** train–val split (`random_state = 42`)
- Drop `SK_ID_CURR`; keep IDs only for the submission file
- Mean imputation for numeric NaN; one-hot encode 16 categoricals → **244 features**
- `StandardScaler` fitted on train only (essential for LR / MLP)

Tuning & validation

- `GridSearchCV` with `PredefinedSplit` (single train→val fold, no random k-fold)
- Scoring: `roc_auc`; best parameter setting picked per model
- CV CSVs saved to `result/home_credit/cv_results/GridSearchCV/`

Classic ML Models

Model	Hyperparameters tuned	Val AUC
Logistic Regression	`C` $\in \{0.01, 0.1, 1, 10\}$, `tol` $\in \{10^{-5}, 10^{-4}, 10^{-3}\}$, `class_weight=balanced`	0.7487
Random Forest	`n_estimators` $\in \{100, 200\}$, `min_samples_split` $\in \{2, 20, 100\}$, `min_samples_leaf` $\in \{1, 20, 100\}$, `class_weight=balanced`	0.7470
HistGradientBoosting	`learning_rate` $\in \{0.01, 0.05, 0.1\}$, `max_iter` $\in \{100, 200\}$, `min_samples_leaf` $\in \{20, 100\}$	**0.7595**

Findings

- **HistGradientBoosting** is the strongest classic learner — boosted trees usually dominate tabular credit data.
- **Logistic Regression** (linear) with `class_weight=balanced` is competitive — most signal is linear in `EXT_SOURCE_*`.
- **Random Forest** overfits aggressively (train AUC 0.96 vs val 0.747) when `min_samples_leaf=1`; the regularized config wins.

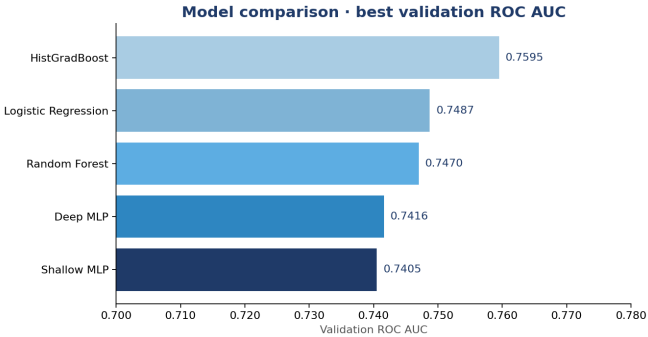
Neural Network Models

Architecture	Hyperparameters tuned	Val AUC
Shallow MLP – hidden $(64,)$	`alpha` $\in \{10^{-4}, 10^{-3}\}$, `learning_rate_init` $\in \{0.001, 0.005\}$, ReLU, Adam, early-stopping	0.7405
Deep MLP – hidden $(128, 64, 32)$	same grid as above	**0.7416**

Findings

- **Best for both:** $\alpha = 10^{-3}$, `lr_init` $= 0.005$ — stronger L2 regularization wins.
- **Depth doesn't help:** 0.7405 vs 0.7416 — a 0.001 difference, well within noise; tripling parameters bought nothing.
- Both NNs land slightly *below* the classic shallow learners — the bottleneck is the feature set, not the model class.

Combined Leaderboard

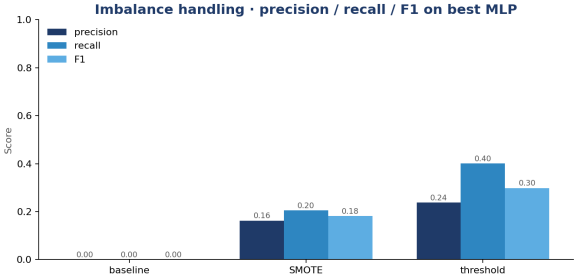


Model	Val AUC
HistGradBoost	**0.7595**
Logistic Reg.	0.7487
Random Forest	0.7470
Deep MLP	0.7416
Shallow MLP	0.7405

Tree boosting wins by $\sim$1 AUC point — consistent with the literature on tabular data.

Imbalance Handling on the Best MLP

Method	Threshold	ROC AUC	PR AUC	Precision	Recall	F1
Baseline	0.500	0.745	0.222	0.00	0.00	0.00
SMOTE	0.500	**0.648**	0.134	0.16	0.21	0.18
Threshold tuning ⋆	0.164	0.745	0.222	0.24	**0.40**	**0.30**



Headline

- Baseline: AUC 0.745 but recall **0%** — no prediction crosses 0.5.
- SMOTE **hurts**: AUC dropped 0.10.
- Threshold tuning at 0.164 lifts recall to **40%**.

Choosing the threshold matters

Conclusions

What we learned

- **HistGradientBoosting** wins overall at val ROC AUC $= 0.7595$ — tree boosting takes the lead by $\sim$1 AUC point.
- **Neural networks** tie LR and RF — depth alone does not unlock new performance on this tabular data.
- **Threshold tuning** on the MLP lifts recall from 0% to 40%; SMOTE hurt AUC.

Limitations

- Only main `application_*.csv` tables — auxiliary tables typically add several AUC points.
- `PredefinedSplit` is fast but optimistic; full k-fold CV would tighten confidence intervals.

Future work

Thank You

Questions?

Recording «INSERT RECORDING LINK BEFORE SUBMISSION»

Code github.com/ymw0414/loan-default-imbalance-classification

Team Minwoo Yoo · Nathaniel Badalov